

# COMP5405 26s1 Final Practice

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## W1 Unit Introduction and Media Computing Map

### Q1. Short Answer

Define digital media computing in terms of a full processing pipeline. Include at least five stages from acquisition to delivery.

### Q2. MCQ

Which answer best describes why an exam answer should state assumptions in a media-size calculation?

- (A) It replaces the need for unit conversion.
- (B) Missing parameters such as frame rate or duration can change the numeric answer.
- (C) It guarantees the data are compressed.
- (D) It makes chrominance subsampling unnecessary.

### Q3. True/False

A method that improves visual appearance is always an image-understanding method. Justify your answer.

### Q4. Table/Diagram

Draw or describe a compact table mapping four media types (audio, image, video, graphics/3D) to their basic data representation and one common processing task.

### Q5. Short Answer

A phone camera app must improve lecture-board photos. Identify three user-visible problems and the technical cause of each.

### Q6. MCQ

A project brief asks for “detect equations and diagrams in board photos”. Which output is most directly required?

- (A) A compression ratio only.
- (B) A binary or labelled region output, such as boxes or masks.
- (C) A colour model conversion only.
- (D) A single audio sampling frequency.

### Q7. Short Answer

Explain why evaluation should include both a quantitative metric and qualitative inspection for a media application.

### Q8. Long Question

Design a high-level processing pipeline for a classroom note-capture tool. Your answer must include acquisition, preprocessing, understanding, storage, and evaluation.

### Q9. True/False

Compressing a board photo heavily before perspective correction and text detection can make later understanding less reliable. Explain.

### Q10. Short Answer

Give two ethical or deployment risks in a digital media application and one mitigation for each.



## W2 Digital Media Basics

### Q1. Calculation Semester 1 2022 Final (modified)

A 20-second RGB video has resolution  $640 \times 480$ , 25 fps, and 8 bits per colour channel. Calculate the uncompressed bytes and MiB. Show all working.

### Q2. Calculation

A stereo audio clip is sampled at 48 kHz with 16-bit samples for 75 seconds. Calculate the raw PCM storage in bytes and MiB.

### Q3. Short Answer

Explain sampling, quantisation, bit depth, and encoding. Use one image or audio example.

### Q4. MCQ

What is the main cause of aliasing?

- (A) Sampling below the rate needed for the signal frequency content.
- (B) Using too many quantisation levels.
- (C) Using a lossless container.
- (D) Increasing spatial resolution.

### Q5. Short Answer

A question gives image width, height, and bit depth but no frame rate or duration. What can and cannot be calculated for a video?

### Q6. Calculation

For an 8-bit uniform quantiser over intensity range 0 to 255, how many levels are available? What is the approximate quantisation step if levels are integer intensities?

### Q7. True/False

Increasing bit depth always increases spatial resolution. Justify your answer.

### Q8. Short Answer

Explain why the human visual system motivates separating luminance from chrominance in video systems.

### Q9. MCQ

Which raw format has the largest bytes per pixel at 8 bits per stored component?

- (A) YCbCr 4:2:0.
- (B) YCbCr 4:2:2.
- (C) RGB 4:4:4.
- (D) 8-bit grayscale.

### Q10. Long Question

A student estimates a video size without stating whether MB means  $10^6$  bytes or MiB means  $2^{20}$  bytes. Explain why this matters and how to present a robust answer.



## W3 Digital Image Processing I

### Q1. Calculation Semester 1 2023 Final (modified)

A grayscale image uses intensity levels 0, 1, 2, and 3 with counts 1, 3, 2, and 2 in an 8-pixel image. Perform histogram equalisation using  $s_k = \text{round}((L-1)CDF(r_k))$ , where  $L = 4$ .

### Q2. Short Answer

Explain the difference between point processing and neighbourhood processing. Give one example of each.

### Q3. Calculation

For the  $3 \times 3$  neighbourhood  $\begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix}$ , apply the Sobel  $G_x$  mask  $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$  as written.

### Q4. MCQ

Histogram equalisation primarily aims to:

- (A) Estimate object bounding boxes.
- (B) Redistribute intensities to improve contrast.
- (C) Reduce audio sampling rate.
- (D) Encode motion vectors.

### Q5. Short Answer

Why can histogram equalisation amplify noise? Mention one safer alternative for uneven illumination.

### Q6. True/False

A constant image patch gives a zero response under the Sobel masks. Explain.

### Q7. Calculation

Apply a threshold of 100 to the row 80, 100, 120, 60, using 1 for values  $\geq 100$  and 0 otherwise.

### Q8. Table/Diagram

Create a three-row table comparing mean filtering, median filtering, and Sobel filtering by purpose and one trade-off.

### Q9. Short Answer

Explain the sign and magnitude of a Sobel  $G_x$  response for a vertical dark-to-bright edge.

### Q10. Long Question

A low-light image is noisy and low contrast. Propose a preprocessing sequence before text detection and justify each step.





## W4 Digital Image Processing II

### Q1. Short Answer Semester 1 2022 Final (modified)

Compare RGB and YUV/YCbCr colour representations. Explain why YUV/YCbCr is commonly used for video.

### Q2. Calculation

For a  $1920 \times 1080$  8-bit YCbCr 4:2:0 frame, calculate bytes per frame.

### Q3. Short Answer

Explain 4:4:4, 4:2:2, and 4:2:0 in terms of luminance and chrominance samples.

### Q4. MCQ

Which method is most directly suited to correcting a planar whiteboard photographed from an angle?

- (A) Homography-based perspective rectification.
- (B) Entropy coding only.
- (C) Audio quantisation.
- (D) Non-maximum suppression.

### Q5. Short Answer

Compare nearest-neighbour, bilinear, and bicubic interpolation in terms of speed and visual quality.

### Q6. True/False

Erosion expands bright foreground regions. Correct the statement if false.

### Q7. Table/Diagram

Give a small table of erosion, dilation, opening, and closing, with one practical use for each.

### Q8. Short Answer

When does global thresholding fail, and why can adaptive thresholding help?

### Q9. Calculation

An RGB image has  $800 \times 600$  pixels and 8 bits per channel. Calculate raw storage in bytes.

### Q10. Long Question

Design a pipeline to turn a skewed, shadowed photo of handwritten notes into a clean binary image. Include at least four processing steps.



## W5 Digital Image Understanding I

### Q1. Short Answer Semester 1 2022 Final (modified)

Describe the Viola-Jones detector. Include Haar-like features, integral images, boosting, and cascade structure.

### Q2. Short Answer

Distinguish image classification, object detection, semantic segmentation, and instance segmentation.

### Q3. MCQ

Which metric directly measures overlap between a predicted box and a ground-truth box?

- (A) IoU.
- (B) Sampling rate.
- (C) Bit depth.
- (D) Gamma.

### Q4. Calculation

A detector gives 40 true positives, 10 false positives, and 20 false negatives. Compute precision and recall.

### Q5. Short Answer

Explain why train/validation/test leakage can produce over-optimistic object-detection results.

### Q6. Table/Diagram

Make a two-column table comparing hand-crafted features and learned CNN features.

### Q7. True/False

High precision always implies high recall. Explain using false positives and false negatives.

### Q8. Short Answer

Give three data augmentation strategies useful for recyclable-object detection on a conveyor belt.

### Q9. MCQ

A confusion matrix is most useful for:

- (A) Seeing which classes are confused with which other classes.
- (B) Calculating raw video bytes directly.
- (C) Removing chroma samples.
- (D) Choosing a DCT block size only.

### Q10. Long Question

Propose an evaluation plan for detecting fruits under different lighting, backgrounds, and orientations. Include dataset split, metrics, and failure analysis.



## W6 Digital Image Understanding II

### Q1. Short Answer Semester 1 2024 Final (modified)

Explain the main pipeline of R-CNN and identify why it is computationally expensive.

### Q2. Short Answer Semester 1 2024 Final (modified)

Explain how Fast R-CNN improves on R-CNN and what computation is shared.

### Q3. MCQ

Which statement is the best critique of "Fast R-CNN is always the best detector"?

- (A) It ignores deployment constraints such as latency, data, hardware, and robustness.
- (B) It is true for every mobile and embedded setting.
- (C) It means evaluation is unnecessary.
- (D) It proves colour conversion is impossible.

### Q4. Short Answer

Compare two-stage and one-stage object detectors in terms of accuracy and speed.

### Q5. Table/Diagram

Draw a compact flow diagram for Fast R-CNN from image to class and bounding-box outputs.

### Q6. Short Answer

What is non-maximum suppression, and how can its IoU threshold affect duplicate or missed detections?

### Q7. Calculation

A predicted box has area 100, a ground-truth box has area 120, and their intersection area is 60. Compute IoU.

### Q8. Short Answer

Explain why bounding-box regression is useful after generating region proposals.

### Q9. True/False

A detector with high mAP on a test set can still fail after deployment. Give two reasons.

### Q10. Long Question

Choose a detector family for a real-time recycling robot. Justify the choice using accuracy, latency, data requirements, and hardware constraints.





## W7 Video Processing

### Q1. Calculation Semester 1 2024 Final (modified)

A  $720 \times 576$  8-bit YCbCr 4:2:2 video runs at 25 fps for 40 seconds. Calculate uncompressed bytes and MiB.

### Q2. Short Answer

Define frame, shot, scene, and keyframe. Explain how they relate in video summarisation.

### Q3. Calculation

For a block-matching candidate, compute SAD between rows 10, 20, 30 and 12, 18, 29.

### Q4. MCQ

Motion estimation mainly exploits which redundancy?

- (A) Temporal redundancy.
- (B) Audio channel redundancy only.
- (C) A larger bit depth.
- (D) CMYK conversion.

### Q5. Short Answer

Explain motion compensation using motion vectors and prediction residuals.

### Q6. True/False

A hard shot cut and a gradual dissolve are equally easy to detect with a single-frame histogram difference. Explain.

### Q7. Table/Diagram

Create a table listing three video-analysis tasks and one cue or feature for each.

### Q8. Short Answer

Why can camera motion make object-motion estimation harder? Give one mitigation.

### Q9. MCQ

Which condition most likely breaks brightness-constancy assumptions in optical flow?

- (A) Sudden lighting change.
- (B) Exact repeated static frames.
- (C) Correct unit conversion.
- (D) Lossless storage.

### Q10. Long Question

Design a short-video mode for a board-capture app. Include stabilisation or motion ideas, compression, and one risk of careless temporal processing.



## W8 Graphics and Animation

### Q1. Short Answer

Describe the graphics pipeline from model coordinates to shaded pixels.

### Q2. MCQ

Which representation is naturally resolution-independent for simple drawings and typography?

- (A) Vector graphics.
- (B) Raw PCM audio.
- (C) A motion vector only.
- (D) An 8-bit histogram.

### Q3. Calculation

A mesh has 2000 triangles. If each triangle is rendered independently with 3 vertices and each vertex stores 3 coordinates as 32-bit floats, what is the coordinate storage before indexing, in bytes?

### Q4. Short Answer

Explain homogeneous coordinates and why they are useful for translation and perspective projection.

### Q5. Table/Diagram

Make a compact diagram of model, view, projection, rasterisation, and shading stages.

### Q6. Short Answer

Compare keyframe animation, physics-based animation, and motion capture.

### Q7. True/False

Increasing mesh detail always improves an interactive AR application. Explain using performance constraints.

### Q8. Short Answer

Explain diffuse and specular reflection at a high level.

### Q9. MCQ

The z-buffer is primarily used to:

- (A) Resolve visibility among surfaces.
- (B) Equalise an image histogram.
- (C) Compute audio bit depth.
- (D) Select chroma samples.

### Q10. Long Question

A mobile AR app must overlay arrows on a live camera feed. Discuss registration, tracking, latency, and rendering trade-offs.





## W9 Visual Computing Advanced I

### Q1. Short Answer Semester 1 2024 Final (modified)

Describe seam carving for content-aware image resizing. Include energy, seam selection, and one advantage over standard resizing.

### Q2. Calculation

For a vertical seam DP row, previous cumulative costs are 5, 8, and 4 for the three possible parent cells, and current energy is 7. What cumulative cost is assigned to the current cell?

### Q3. MCQ

Seam carving is least suitable when:

- (A) Important straight structures or faces would be distorted.
- (B) The background is low-detail texture.
- (C) The energy map clearly separates content.
- (D) Protected regions are correctly masked.

### Q4. Short Answer

Explain how a protect mask or remove mask can change seam-carving behaviour.

### Q5. Short Answer

Outline an OCR pipeline for a board image from capture to recognised text.

### Q6. True/False

Super-resolution always recovers true physical detail that was present in the scene. Explain.

### Q7. Table/Diagram

Compare interpolation-based, reconstruction-based, and deep-learning-based super-resolution.

### Q8. Short Answer

Explain HDR image fusion and one failure case.

### Q9. MCQ

Which metric is most directly tied to OCR output quality?

- (A) Character error rate.
- (B) Frames per second only.
- (C) DCT coefficient count only.
- (D) Chroma sampling ratio only.

### Q10. Long Question

Design a visual-computing pipeline for scanning a poster under glare and perspective distortion. Include enhancement, geometry, understanding, and evaluation.



## W10 Visual Computing Advanced II

### Q1. Short Answer

Explain what makes a visual computing system “advanced” beyond applying a single image filter.

### Q2. MCQ

A hybrid approach is often useful because it:

- (A) Combines reliable classical preprocessing with learned semantic recognition.
- (B) Avoids evaluation entirely.
- (C) Uses only audio sampling.
- (D) Removes all deployment constraints.

### Q3. Short Answer

For detecting diagrams, equations, and handwriting, compare a classical, deep-learning, and hybrid approach.

### Q4. Table/Diagram

Create a table with criteria: data requirements, robustness, computation, interpretability, and deployment. Fill it for classical vs deep learning.

### Q5. True/False

A model trained on clean scanned documents will necessarily work on low-light mobile photos. Explain.

### Q6. Short Answer

Explain domain shift and give two ways to reduce its effect.

### Q7. MCQ

Which evaluation pair is strongest for a note-capture feature?

- (A) OCR accuracy plus human readability inspection.
- (B) File name length plus page count only.
- (C) Bit depth only.
- (D) Number of lectures only.

### Q8. Short Answer

Give three deployment constraints for mobile visual computing and how they affect algorithm choice.

### Q9. Calculation

A detector processes one frame in 80 ms on a phone. What is the maximum fps if processing is sequential and no other overhead is counted?

### Q10. Long Question

Design an “Exam Board Capture” mode. Use at least six techniques from the course and justify each under low light, blur, skew, shadows, and limited storage.



## W11 Media Compression

### Q1. Short Answer Semester 1 2023 Final (modified)

Compare JPEG and JPEG2000. Mention transform type, scalability, typical artefacts or complexity, and one suitable application.

### Q2. Short Answer

List and explain spatial, temporal, coding, and perceptual redundancy. Give one technique exploiting each.

### Q3. MCQ

In baseline JPEG, quantisation occurs after:

- (A) DCT transformation of image blocks.
- (B) Object detection NMS.
- (C) Audio sampling only.
- (D) Homography estimation only.

### Q4. Table/Diagram

Draw or list the main JPEG pipeline from colour conversion to entropy coding.

### Q5. Calculation

A raw image is 6 MiB and compressed to 1.5 MiB. What is the compression ratio as raw/compressed?

### Q6. Short Answer

Explain why too much quantisation creates visible artefacts. Name one JPEG artefact.

### Q7. True/False

Lossless compression can change pixel values if the decoder is correct. Explain.

### Q8. Short Answer

Explain why inter-frame video compression is efficient but can propagate errors.

### Q9. MCQ

Which metric is computed from MSE and maximum possible pixel value?

- (A) PSNR.
- (B) IoU.
- (C) Recall.
- (D) Sampling frequency.

### Q10. Long Question

Choose compression settings for storing board-capture images and short clips. Discuss readability, file size, artefacts, and task metrics.





## W12 Project Presentations and Synthesis

### Q1. Short Answer

State a reusable structure for answering an integrated media-system design question.

### Q2. MCQ

Which answer is strongest in a design question?

- (A) A list of algorithms without explanation.
- (B) A pipeline where each method is tied to a problem, trade-off, and metric.
- (C) A single file-size number only.
- (D) A claim that deep learning always solves the task.

### Q3. Calculation

For a  $720 \times 576$  4:2:2 frame at 8 bits/sample, compute bytes per frame and compare it to RGB24 bytes per frame as a ratio.

### Q4. Short Answer

Explain how to handle a question that intentionally omits necessary calculation parameters.

### Q5. Table/Diagram

Create a table mapping four system constraints (latency, storage, lighting, privacy) to design responses.

### Q6. True/False

A qualitative user study is useless if quantitative metrics are available. Explain.

### Q7. Short Answer

Give two ways dataset bias can affect a media-recognition system and two mitigations.

### Q8. MCQ

Which is a good final-exam phrase for a conditional recommendation?

- (A) This method is suitable because it targets the stated problem under the deployment constraint, but it may fail when the assumption breaks.
- (B) This method is always best.
- (C) The formula is not needed.
- (D) Evaluation is unnecessary.

### Q9. Short Answer

Propose one quantitative and one qualitative evaluation for an OCR-based note-capture system.

### Q10. Long Question

Write a full design answer for an innovative camera-app feature that improves student note capture. Include at least five course techniques, trade-offs, and evaluation.



## Answer Key

### W1

- Q1. Acquisition, digitisation, representation, preprocessing, analysis/understanding, compression/storage, evaluation.
- Q2. B. Missing parameters directly change storage or rate.
- Q3. False. Enhancement changes appearance; understanding infers semantic content such as objects or text.
- Q4. Audio: samples/waveform, denoise or speech recognition. Image: pixels, enhancement/detection. Video: frames over time, motion/shot analysis. Graphics/3D: geometry/material/camera, rendering/animation.
- Q5. Examples: blur from motion/defocus, low contrast from poor illumination, skew from perspective, colour cast from lighting, shadows from occlusion.
- Q6. B. Region outputs such as boxes/masks are required.
- Q7. Metrics give repeatable evidence; inspection catches artefacts and user-facing failures not captured by one number.
- Q8. A strong answer includes capture, white balance/denoise/CLAHE, homography, segmentation/OCR/detection, compression, and metrics such as OCR accuracy/readability.
- Q9. True. Heavy compression can introduce blocking/blur/ringing artefacts before geometry correction or OCR, so detection and recognition may become less reliable.
- Q10. Privacy/consent, dataset bias, misleading enhancement, compute/energy. Mitigate by minimisation, consent, diverse data, audit, and user warnings.

### W2

- Q1. Frames =  $25 \cdot 20 = 500$ . Bytes/frame =  $640 \cdot 480 \cdot 3 = 921600$ . Total  $460800000 \text{ B} \approx 439.45 \text{ MiB}$ .
- Q2.  $48000 \cdot 75 \cdot 2 \cdot 2 = 14400000 \text{ B} \approx 13.73 \text{ MiB}$ .
- Q3. Sampling discretises time/space; quantisation discretises amplitude; bit depth sets levels; encoding stores bits/container.
- Q4. A.
- Q5. Can calculate bytes per frame only; total video size needs number of frames from fps and duration.
- Q6.  $2^8 = 256$  levels; step is one integer intensity level across 0–255.
- Q7. False. Bit depth affects amplitude precision, not spatial pixel count.
- Q8. Luminance carries much perceived detail; chroma can be sampled more coarsely with less visible damage.
- Q9. C. RGB 4:4:4 uses 3 bytes/pixel at 8 bits/channel.
- Q10. State whether using decimal MB or binary MiB; show bytes first and convert explicitly.

### W3

- Q1. PMF/probabilities  $1/8, 3/8, 2/8, 2/8$ ; CDF  $0.125, 0.5, 0.75, 1$ ; map 0, 2, 2, 3 using round  $3CDF$ .
- Q2. Point processing maps one pixel independently, e.g. gamma. Neighbourhood processing uses surrounding pixels, e.g. convolution.
- Q3.  $G_x = (-10 + 80) + (-20 + 160) + (-10 + 80) = 280$ .
- Q4. B.
- Q5. It stretches intensity differences including noise. CLAHE with clipping or denoising first can help.
- Q6. True. Positive and negative Sobel weights sum to zero on a constant patch.
- Q7. 0, 1, 1, 0.
- Q8. Mean smooths noise but blurs; median removes impulse noise while preserving edges; Sobel estimates gradients but is noise-sensitive.
- Q9. Magnitude gives edge strength; sign depends on mask convention and dark-to-bright direction.
- Q10. Denoise, contrast enhancement, possible illumination correction, then threshold/OCR/detection; mention trade-offs such as blur.

### W4

- Q1. RGB stores device colour channels; YUV/YCbCr separates luminance from chrominance, enabling chroma subsampling and perceptual compression.
- Q2.  $1920 \cdot 1080 \cdot 1.5 = 3110400 \text{ B}$ .
- Q3. 4:4:4 stores full luma and chroma; 4:2:2 halves horizontal chroma; 4:2:0 halves chroma horizontally and vertically.
- Q4. A.
- Q5. Nearest is fastest/blocky; bilinear smoother; bicubic smoother/sharper but costlier.
- Q6. False. Erosion shrinks bright foreground; dilation expands it.





- Q7. Erosion removes specks; dilation fills/bridges; opening removes small foreground; closing fills small holes.  
 Q8. Global threshold fails under uneven lighting/shadows; adaptive uses local thresholds.  
 Q9.  $800 \cdot 600 \cdot 3 = 1440000$  B.  
 Q10. Homography rectification, denoise, illumination correction/CLAHE, adaptive thresholding, morphology, va

## W5

- Q1. Haar-like features computed quickly with integral images; AdaBoost selects features; cascade rejects negatives early.  
 Q2. Classification labels whole image; detection gives boxes/classes; semantic segmentation labels pixels by class; instance segmentation separates objects.  
 Q3. A.  
 Q4.  $\text{Precision} = 40/(40 + 10) = 0.8$ .  $\text{Recall} = 40/(40 + 20) = 0.667$ .  
 Q5. Leakage puts near-duplicates/scenes in test, hiding poor generalisation.  
 Q6. Hand-crafted features are designed and interpretable but limited; CNN features are learned and powerful but data/compute hungry.  
 Q7. False. High precision can coexist with low recall if many objects are missed.  
 Q8. Lighting, blur, occlusion, scale/crop, rotation, background variation.  
 Q9. A.  
 Q10. Include split, no leakage, annotation checks, precision/recall/mAP, per-class matrix, lighting/orientation failure cases.

## W6

- Q1. R-CNN proposes regions, warps each, runs CNN per proposal, classifies and refines boxes; repeated CNN work is expensive.  
 Q2. Fast R-CNN computes one shared feature map, pools RoI features, then classifies/regresses boxes.  
 Q3. A.  
 Q4. Two-stage often accurate but slower; one-stage usually faster but may trade localisation/rare-object accuracy.  
 Q5. Image  $\rightarrow$  CNN feature map  $\rightarrow$  RoI pooling/align  $\rightarrow$  class scores and bbox regression.  
 Q6. NMS removes duplicate boxes; low threshold may suppress close objects, high threshold may keep duplicates.  
 Q7.  $\text{IoU} = 60/(100 + 120 - 60) = 60/160 = 0.375$ .  
 Q8. It corrects proposal location/size to better match objects.  
 Q9. True. Domain shift, rare classes, lighting, occlusion, annotation mismatch, or hardware latency can cause failure.  
 Q10. A justified YOLO/SSD-like choice for real-time, or Faster/Fast R-CNN if accuracy dominates, with data/hardware trade-offs.

## W7

- Q1.  $4:2:2 = 2$  B/pixel. Frame =  $720 \cdot 576 \cdot 2 = 829440$  B. Frames = 1000. Total  $829440000$  B  $\approx 791.02$  MiB.  
 Q2. Frame is one image; shot is continuous take; scene groups shots; keyframe represents a shot.  
 Q3.  $\text{SAD} = |10 - 12| + |20 - 18| + |30 - 29| = 5$ .  
 Q4. A.  
 Q5. Use motion vectors to predict current frame from reference frame; encode residual differences.  
 Q6. False. Gradual dissolves may need temporal windows; single-frame difference suits hard cuts better.  
 Q7. Shot detection: hist diff; tracking: optical flow; compression: motion estimation; summary: keyframes.  
 Q8. Camera motion creates global apparent motion; mitigate by stabilisation or global motion compensation.  
 Q9. A.  
 Q10. Include stabilisation, motion estimation/keyframes, inter-frame compression, OCR/readability preservation, and risks such as ghosting/error propagation.

## W8

- Q1. Model coordinates, world/view transform, projection, rasterisation, visibility, shading, display.  
 Q2. A.  
 Q3.  $2000 \cdot 3 \cdot 3 \cdot 4 = 72000$  B.  
 Q4. Homogeneous coordinates represent translation/projective transforms as matrix multiplication and allow perspective divide.  
 Q5. Model  $\rightarrow$  view  $\rightarrow$  projection  $\rightarrow$  rasterise  $\rightarrow$  shade.  
 Q6. Keyframes interpolate artist-set poses; physics sim uses forces; mocap records real motion.  
 Q7. False. More detail can reduce fps, increase memory, and harm AR latency.





Q8. Diffuse depends on angle to light; specular models shiny view-dependent highlights.

Q9. A.

Q10. Need accurate registration/tracking, calibrated camera/pose, low latency, efficient rendering, and graceful degradation.

## W9

Q1. Compute energy, find low-energy connected seam via DP, remove/insert seam; preserves salient content better than uniform scaling.

Q2.  $7 + \min(5, 8, 4) = 11$ .

Q3. A.

Q4. Protect masks raise energy; remove masks lower energy in regions to remove.

Q5. Rectify, denoise/enhance, binarise/segment, recognise characters/words, post-correct, evaluate CER/WER.

Q6. False. SR may infer or hallucinate plausible detail rather than recover ground truth.

Q7. Interpolation is simple; reconstruction uses multiple observations/priors; DL learns mapping but needs data and can hallucinate.

Q8. Fuse multiple exposures after alignment and tone mapping; motion causes ghosting.

Q9. A.

Q10. Include glare reduction/HDR, homography, denoise/CLAHE, segmentation/detection, compression, readability/OCR metrics.

## W10

Q1. It integrates representation, enhancement, geometry, recognition, constraints, and evaluation rather than one isolated filter.

Q2. A.

Q3. Classical: interpretable/low data but brittle; DL: robust if trained but data/compute hungry; hybrid balances both.

Q4. Classical lower data/compute and interpretable; DL higher robustness to variation but needs labels/compute; deployment depends hardware.

Q5. False. Domain shift from scan to mobile photos can break assumptions.

Q6. Domain shift is train/test distribution mismatch; reduce with representative data, augmentation, adaptation, validation.

Q7. A.

Q8. Battery, memory, latency, privacy, offline operation, model size affect algorithm and compression choices.

Q9.  $1/0.08 = 12.5$  fps maximum.

Q10. Strong answer uses colour conversion, denoise, white balance, homography, CLAHE/adaptive threshold, OCR/detection, compression, metrics.

## W11

Q1. JPEG uses block DCT and is common/simple but blocky; JPEG2000 uses wavelets, supports scalability/ROI, often better low-rate but complex.

Q2. Spatial: DCT/intra; temporal: motion; coding: entropy; perceptual: chroma/quantisation.

Q3. A.

Q4. RGB to YCbCr, subsample, block, DCT, quantise, zigzag/RLE, entropy code.

Q5.  $6/1.5 = 4 : 1$ .

Q6. Coarse quantisation discards coefficients; artefacts include blocking, ringing, blur.

Q7. False. Correct lossless decoding restores exact pixels.

Q8. Efficient because frames are similar; errors may affect predicted future frames.

Q9. A.

Q10. Use PNG/lossless or high-quality JPEG for text-heavy images, video codec for clips, evaluate OCR/readability plus size.

## W12

Q1. Diagnose, choose representation, pipeline, justify each step, state assumptions/trade-offs, evaluate.

Q2. B.

Q3.  $4:2:2$  frame =  $720 \cdot 576 \cdot 2 = 829440$  B;  $RGB24 = 720 \cdot 576 \cdot 3 = 1244160$  B; ratio  $RGB/4:2:2 = 1.5$ .

Q4. State missing info, avoid hidden defaults, give symbolic formula or explicit assumption.

Q5. Latency: fast detector/model compression; storage: compression; lighting: enhancement/HDR; privacy: local processing/minimisation.

Q6. False. Qualitative studies reveal user-facing artefacts and usability issues.



Q7. Bias can reduce performance on unseen lighting, handwriting, devices, or languages; mitigate with diverse augmentation, and validation.

Q8. A.

Q9. Quantitative: OCR CER/WER or detection mAP; qualitative: human readability/user study/failure gallery.

Q10. Full answer should tie each method to low light, blur, skew, colour/shadows, text/diagram detection, storage.